Improvement in Modulation Bandwidth of Micro-LED Arrays Based on Low-Temperature-Interlayer Approach

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***Abstract*— To promote the modulation bandwidth of GaN-based blue micro-light emitting diodes (μ-LEDs) for visible light communication, the multiple quantum well active region has been improved via approaches based on low-temperature (LT) GaN insertion layer in barriers. μ-LED arrays with 10 × 10 pixels have been fabricated. The device using LT-GaN interlayer has an output power of 2.14 mW and a modulation bandwidth of 220 MHz, with an enhancement factor of 2.5 and 1.7, respectively, as compared with the device without LT-GaN interlayer. It is believed that these improvements are a result of the suppression of strain-related quantum-confined Stark effect and defects/dislocations by LT-GaN interlayer based optimization. The LT-GaN interlayer not only facilitates a tensile strain in barriers, making it more lattice-compatible with InGaN well to reduce the strain there, but also has an effect on blocking the propagation of threading dislocation into upper layers. Our work provides an effective approach to enhance the modulation bandwidth of III-nitride μ-LEDs using LT interlayer.**

Manuscript received May 6, 2022; revised May 29, 2022; accepted June 5, 2022. Date of publication June 8, 2022; date of current version June 16, 2022. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFA0716400, in part by the Beijing Natural Science Foundation under Grant Z200004, in part by the Beijing Outstanding Young Scientist Program under Grant BJJWZYJH0120191000103, in part by the National Natural Science Foundation of China under Grant 61734001, in part by the Key-Area Research and Development Program of Guangdong Province under Grant 2019B010132001, and in part by the Basic Research Foundation under Grant 2019B1515120091, and Grant 2021B1515120022. *(Corresponding authors: Renchun Tao; Xinqiang Wang.)*

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Color versions of one or more figures in this letter are available at https://doi.org/10.1109/LPT.2022.3181275.

Digital Object Identifier 10.1109/LPT.2022.3181275

***Index Terms*— Multiple quantum wells, low-temperature interlayer, blue micro-light emitting diodes, modulation bandwidth.**

I. Introduction

In RECENT years, GaN-based micro-light emitting diodes (μ-LEDs) have attracted extensive attentions due to the excellent characteristics of self-luminous, wide color gamut, micro-size, low-power consumption, high injection current density, and so on [1]–[3], which have found potential applications in advanced display for virtual reality (VR) and augmented reality (AR), visible light communication (VLC), automobile monitor, and wearable devices [4]. However, for VLC application, the modulation bandwidth of GaN-based μ-LEDs is still limited. The traditional bulky device with size of 300 μm$\sim$1 mm is usually limited to 10$\sim$20 MHz due to a large RC delay. By scaling down to micro-size (1$\sim$100 μm), the RC delay effects can be significantly reduced and the resulted μ-LEDs have shown enhanced modulation bandwidth. Herein, one of the main factors limiting the modulation bandwidth of μ-LEDs is the quantum-confined Stark effect (QCSE) intrinsic to III-nitride nanostructures, where the strong built-in electrical field induced by spontaneous and piezoelectric polarization separates electrons and holes, giving a long carrier lifetime. Thus, according to the roughly inverse proportion relationship between modulation bandwidth (f) and carrier lifetime (τ) of μ-LEDs [5], i.e., $f \propto 1/\tau$, the modulation bandwidth of GaN-based μ-LEDs is still relatively low. In order to increase the modulation bandwidth, the most straightforward approach is to decrease the polarization field [6]. Since the spontaneous polarization is usually fixed for active regions with a target emission wavelength, the regulation of strain-involved piezoelectric polarization becomes an important factor to suppress QCSE and increase the modulation bandwidth of GaN-based μ-LEDs [7].

Several methods have been proposed to ease the strain status in the active region of GaN-based LEDs. For example, incorporating silicon dopants into GaN quantum barrier (QB) [8], replacing GaN barriers with InGaN ones in InGaN multiple quantum wells (MQWs) [9], inserting thin strain-modulation AlGaN/InAlN layers [10], and introducing InGaN/GaN or InGaN/InGaN superlattices (SLs) before depositing active regions [11] have shown to be effective in releasing the strain and suppressing QCSE in InGaN active regions. In particular, the method with insertion of low-temperature (LT) thin layers

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in growing process not only modulates the strain state, but also reduces the defect/dislocation density in later growing layers [12]. It has been used to improve the emission performance of III-nitride active regions. However, to the best of our knowledge, no study with such LT interlayer method has been reported in GaN-based μ-LEDs to suppress QCSE and promote their modulation bandwidth yet.

In this work, the emission property of blue LED structures has been systematically studied based on MQW barriers with LT-GaN interlayer. Then, barrier, well thicknesses, and barrier doping in active region are further optimized. The blue μ-LED arrays using u-$In_{0.18}Ga_{0.82}N$/n-GaN MQWs with LT-GaN interlayer are fabricated, and reference blue μ-LED arrays using traditional u-$In_{0.18}Ga_{0.82}N$/u-GaN MQWs without LT-GaN interlayer are also prepared. Both types of blue μ-LED arrays have 10 × 10 pixels. By employing u-$In_{0.18}Ga_{0.82}N$/n-GaN MQWs with LT-GaN interlayer, the reduced density of defects or dislocations related to V-shaped pits and the suppression of QCSE effectively improve the device's light output power (LOP) to 2.14 mW and the modulation bandwidth to 220 MHz, which shows an enhancement of 2.5 and 1.7 times as compared with the reference device, respectively.

II. Experimental Procedure

2″ c-plane sapphires were used to prepare the LED devices by AIXTRON 19 × 2″ metal-organic chemical vapor deposition (MOCVD). The whole LED epi-structures include15-nm-thick sputtered-AlN buffer, 3-μm-thick u-GaN template, 2-μm-thick n-GaN layer with electron concentration of 1×10^{18} cm^{-3}, 15-nm-thick u-GaN first barrier, 3 periods of InGaN/GaN MQWs, 15-nm-thick u-GaN spacer layer, 20-nm-thick p-$Al_{0.2}Ga_{0.8}N$ electron blocking layer (EBL), 13 periods of p-$Al_{0.2}Ga_{0.8}N$/p-GaN SLs, 60-nm-thick polarization-induced p-type doping of graded AlGaN, and 5-nm-thick p^{++}-GaN with doped-Mg concentration of 3.5×10^{20} cm^{-3}. The hole concentration of p-type layer composed by p-$Al_{0.2}Ga_{0.8}N$/p-GaN SLs, graded p-AlGaN, and p^{++}-GaN was measured to be 1.7×10^{18} cm^{-3}. In order to enhance the device's luminescent property and modulation bandwidth in VLC, MQWs structure was optimized based on LT-GaN interlayer in barriers, barrier doping, barrier and well thickness variation. For the reference device (noted as μ-LED1, Fig. 1(a)), each QW period consists of a 3.5-nm-thick u-$In_{0.18}Ga_{0.82}N$ well, 2-nm-thick u-GaN cap layer, and 10-nm-thick u-GaN barrier. For the target device (noted as μ-LED2, Fig. 1(a)), each QW period includes a 2-nm-thick LT u-GaN interlayer, 3-nm-thick u-$In_{0.18}Ga_{0.82}N$ well, 2-nm-thick u-GaN cap layer, and 15-nm-thick n-GaN barrier with electron concentration of 1.5×10^{17} cm^{-3}. The optimization process of MQW structure for μ-LED2 is discussed below in details.

III. Results and Discussion

For μ-LED1 and μ-LED2 epi-structures, X-ray diffraction (XRD) reciprocal space mappings (RSMs) in asymmetric (10-15) planes were measured to characterize device structure property and strain status. As shown in Fig. 1(b) for μ-LED1

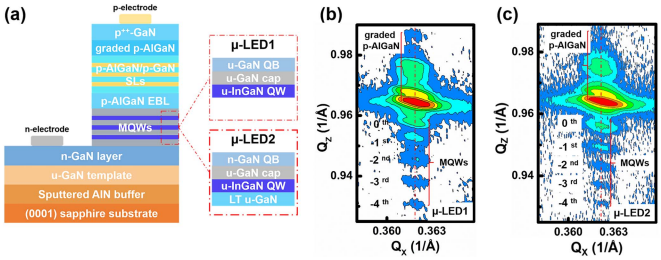

Fig. 1. (a) Schematic diagram of blue μ-LEDs with different MQW structures. The XRD-RSMs for the (10-15) planes of (b) μ-LED1 and (c) μ-LED2.

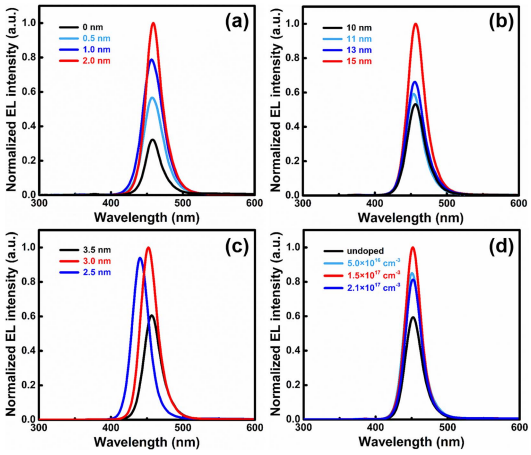

Fig. 2. EL spectra measured at 100 mA with different (a) LT u-GaN thickness, (b) GaN QB thickness, (c) InGaN QW thickness, and (d) GaN QB doping concentration, respectively.

and Fig. 1(c) for μ-LED2, the features of polarization-induced graded p-AlGaN with gradient Al content distribution, and four-order diffraction satellite peaks formed by InGaN/GaN MQWs can be well distinguished. Moreover, it is obvious that both devices are coherently grown, as no obvious relaxation behaviour (red dot line) can be observed.

A. Improvement of Luminescence Property

Based on the MQW structure (3.5 nm for u-$In_{0.18}Ga_{0.82}N$ well, 2 nm for u-GaN cap layer, and 10 nm for u-GaN barrier) of μ-LED1, LT-GaN layers were inserted into MQW barriers, and then the thicknesses of wells and barriers were optimized. Besides, MQW barrier doping was also used to improve luminescent property. For emission property characterization, electroluminescence (EL) measurements were performed for all samples. Firstly, LT-GaN interlayers with different thicknesses were inserted at the bottom of each u-$In_{0.18}Ga_{0.82}N$ well, with a growth temperature of 755 °C and triethylgallium (TEGa) flow rate of 8.1 μmol/min. Their EL spectra are shown in Fig. 2(a), where the peak intensity gradually increases with increasing the thickness of LT-GaN interlayer from 0 to 2 nm. Then, with a LT-GaN of 2-nm-thick, the barrier thickness was varied at 825 °C with trimethylgallium (TMGa) flow rate of 95.5 μmol/min. Fig. 2(b) shows the EL peak intensity increases with barrier thickness increased from 10 to 15 nm. It is possibly because that the carrier penetration probability into barriers is reduced with increasing barrier thickness, so that the radiative recombination rate in the well region increases [13]. Later, with 2-nm-thick LT-GaN interlayers and

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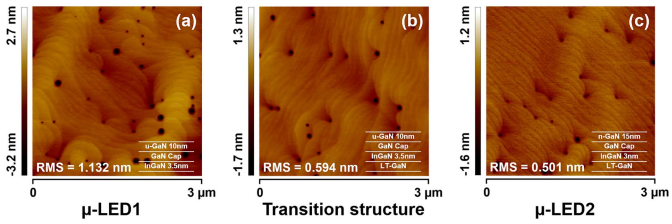

Fig. 3. The representative 3×3 μm^2 AFM micrographs of MQWs surface corresponding to (a) μ-LED1, (b) transition structure, and (c) μ-LED2.

15-nm-thick barrier layers, the well thickness was optimized with their measured EL spectra presented in Fig. 2(c). It is clear that the EL peak intensity significantly increases and then slightly drops with well thickness decreasing from 3.5 to 2.5 nm, which should be due to the enhanced wave-function overlap between electrons and holes as the decreased well thickness tends to push electrons and holes closer [14] and over thinned well that would weaken the confinement ability for carriers and decrease radiative recombination rate. Finally, barrier doping was tried to furthur increase emission intensity. Fig. 2(d) shows the peak intensity firstly increases and then decreases with n-type doping concentration increasing from 0 to 2.1×10^{17} cm^{-3}, and the best doping concentration is 1.5×10^{17} cm^{-3}. It is possibly due to the crystalline quality improvement for barrier and the reduction of compressive stress for well [15] with a relatively low n-type doping concentration. With all the above optimizations, the best device is noted as μ-LED2, which has 2-nm-thick LT-GaN interlayer, 3-nm-thick u-$In_{0.18}Ga_{0.82}N$ well layer, 2-nm-thick u-GaN cap layer, and 15-nm-thick n-GaN barrier layer with a doping concentration of 1.5×10^{17} cm^{-3}.

In order to verify the improvement on surface smoothness and crystalline quality of MQW layers by using LT-interlayer-based optimization process, three additional samples were fabricated with expitaxy terminated after MQW growth: two samples corresponding to μ-LED1 and μ-LED2, and a transition sample with the same MQW structure as μ-LED1 except for the insertion of 2-nm-thick LT-GaN layers. Atomic force microscope (AFM) measurements with a scan area of 3×3 μm^2 were performed with results shown in Fig. 3. The AFM image of MQWs using 2-nm-thick LT-GaN interlayer (Fig. 3(b), transition structure), presents a smooth surface with a root-mean-square (RMS) value of 0.594 nm, much smaller than that of μ-LED1 (1.132 nm). Meanwhile, the density of V-shaped pits related with defects or dislocations [8] in the transition MQW structure with LT interlayer is estimated to 2.8×10^8 cm^{-2}, lower than that of μ-LED1 (5×10^8 cm^{-2}). These results confirm the insertion of LT-GaN layer is conducive to improve surface smoothness and reduce defects or dislocations density of MQW layers. With additional structure optimzation on MQWs with LT interlayers, the surface of MQWs for LED2, is further improved, with RMS decreased to 0.501 nm and the density of defects or dislocations to 2.3×10^8 cm^{-2}.

B. Fabrication and Characterization of Blue μ-LED Arrays

Using LED1 and LED2 epi-structures, μ-LED arrays were fabricated by standard semiconductor processes, with each array including 10×10 pixels, mesas of 20 μm in diameters, and electrodes of 10 μm in diameters. Meanwhile, the period length between two closest pixels is 40 μm, and p-type (n-type) ohmic contact electrodes are Ni/Au (Ti/Al/Ti/Au) metals. The two fabricated array devices are noted as μ-LED1 and μ-LED2, respectively.

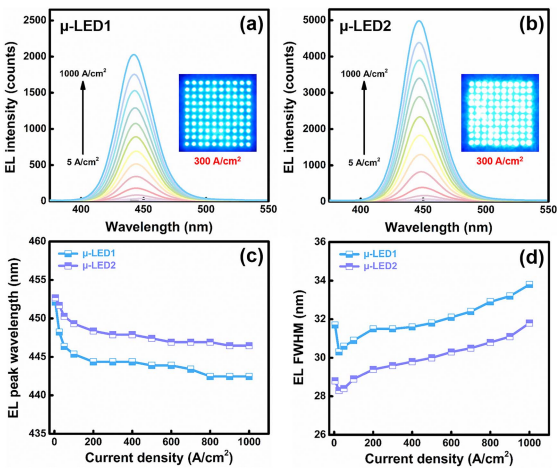

Fig. 4. EL spectra of (a) μ-LED1 and (b) μ-LED2, with injection current density increasing from 5 to 1000 A/cm^2. The insets in (a) and (b) show the plan-view emission images driven at 300 A/cm^2. (c) EL peak wavelength and (d) EL FWHM value of μ-LED1 and μ-LED2 at different injection current densities.

Figs. 4(a) and 4(b) show the room-temperature EL spectra of μ-LED1 and μ-LED2 under different injection current density, respectively. As the injected current density increases from 5 to 1000 A/cm^2, the integrated intensity of EL spectra for both devices gradually enhances, and the integrated EL intensity of μ-LED2 is about 2.5 times in magnitude higher than that of μ-LED1 with the same driven current density of 1000 A/cm^2. The insets in Figs. 4(a) and 4(b) present their emission images driven at 300 A/cm^2, respectively. Their peak wavelength and full width at half maximum (FWHM) value of EL spectra as a function of injection current density are shown in Figs. 4(c) and 4(d), respectively. μ-LED2 exhibits a smaller blue shift of 6.1 nm with wavelength changing from 452.3 to 446.2 nm, which is less than 9.7 nm of μ-LED1 with increasing injection current density from 5 to 1000 A/cm^2. The smaller blue shift in μ-LED2 reveals a suppressed QCSE in MQWs due to the weaker internal polarization-field in active region [16]. Moreover, the FWHM value of μ-LED2 is narrower than that of μ-LED1, which is consistent with the AFM measurements (Fig. 3), where the surface of MQWs corresponding to μ-LED2 is flatter than μ-LED1. For both devices, EL spectra broadening with increasing injection current density are also noted (Fig. 4(d)), possibly due to the competitive mechanism between state-filling effect and heating effect [17].

The LOP of μ-LED1 and μ-LED2 as a function of driven current density is shown in Fig. 5(a). For μ-LED1, the optical output power is 0.13 mW at 25 A/cm^2 and linearly increases to 0.94 mW at 1000 A/cm^2. For μ-LED2, the output power is enhanced from 0.17 to 2.14 mW as current density increasing from 25 to 1000 A/cm^2. At 1000 A/cm^2, the LOP of μ-LED2 (2.14 mW) is about 2.5 times in magnitude higher than that of μ-LED1 with 0.94 mW. The enhanced LOP of μ-LED2 is due to the reduction in the densities of defects or dislocations as

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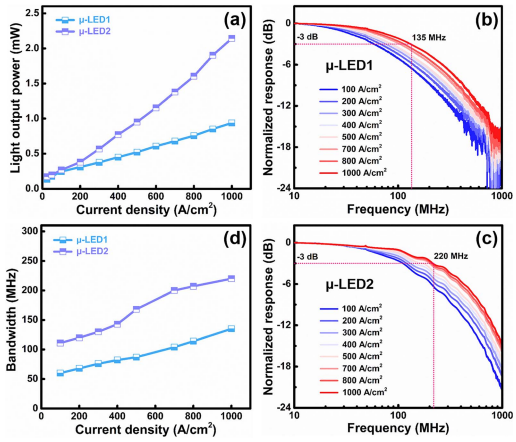

Fig. 5. (a) Light output power versus injection current density of μ-LED1 and μ-LED2. The frequency response curves of (b) μ-LED1 and (c) μ-LED2 at different current density. (d) The -3 dB modulation bandwidth of μ-LED1 and μ-LED2 as a function of current density.

confirmed by AFM measurements (Fig. 3(c)) and the suppression of QCSE revealed by EL measurements (Fig. 4(c)). Subsequently, for VLC application, the modulation bandwidths of μ-LED1 and μ-LED2 were evaluated by a vector network analyzer (Pico VNA 106) to analyse the amplitude-frequency characteristics of both devices. The measured frequency response characteristic curves of μ-LED1 and μ-LED2 are presented in Figs. 5(b) and 5(c), respectively, with injection current density increasing from 100 to 1000 A/cm^2. Fig. 5(d) describes the extracted -3 dB modulation bandwidths of μ-LED1 and μ-LED2 with different injection current density. For μ-LED1, the modulation bandwidth is 60 MHz at 100 A/cm^2 and increases to 135 MHz at 1000 A/cm^2. The modulation bandwidth of μ-LED2 is higher than that of μ-LED1 driven at each current density. At 1000 A/cm^2, μ-LED2 has a modulation bandwidth of 220 MHz, which exhibits an enhancement of 70% as compared with μ-LED1 of 135 MHz.

Both the improved optical output power and modulation bandwidth of μ-LED2 are believed to originate from the LT-GaN interlayer based MQW optimization. Firstly, the involvement of LT-GaN interlayer facilitates a tensile strain in barriers [12], making it more lattice-compatible with InGaN well layers. As a result, the suppressed QCSE due to strain relief in well region would benefit a faster recombination rate and higher optical output power (Fig. 5(a)). Secondly, the LT-GaN interlayer has an effect on blocking the penetration of threading dislocation into upper layers, as confirmed by the AFM images of MQWs for transition structure and μ-LED2 (Figs. 3(b) and 3(c)), which also helps increase device optical output power. Thirdly, the faster recombination rate resulted from the suppression of QCSE would lead to a shorter carrier lifetime. As known that the modulation bandwidth is inversely proportional to the carrier lifetime, μ-LED2 exhibits a much improved modulation bandwidth as compared with μ-LED1 (Fig. 5(d)).

IV. Conclusion

In conclusion, III-nitride MQW structures have been optimized with processes based on LT-GaN interlayer to enhance the modulation bandwidth of blue μ-LED arrays for VLC application. The optimized μ-LED arrays using u-InGaN/n-GaN MQW with LT-GaN interlayer has shown a LOP of 2.14 mW and modulation bandwidth of 220 MHz, which presents an enhancement of 2.5 and 1.7 times as compared with μ-LED arrays without LT-GaN interlayer. The enhanced output power and modulation bandwidth are attributed to the reduction in densities of defects or dislocations and the suppression of QCSE using LT-GaN interlayer based optimization processes in MQWs. These results encourage the implementation of LT interlayer in MQWs of III-nitride μ-LEDs for higher luminescence efficiency and modulation bandwidth.

References

[1] S. Zhang *et al.*, "A high-speed visible light communication system using pairs of micro-size LEDs," *IEEE Photon. Technol. Lett.*, vol. 33, no. 18, pp. 1026–1029, Sep. 15, 2021.

[2] J. P. Huang, Z. L. Hu, X. Gao, Y. Xu, and L. C. Wang, "Unidirectional-emitting GaN-based micro-LED for 3D display," *Opt. Lett.*, vol. 46, no. 14, pp. 3476–3479, Jul. 2021.

[3] E. Xie *et al.*, "Over 10 Gbps VLC for long-distance applications using a GaN-based series-biased micro-LED array," *IEEE Photon. Technol. Lett.*, vol. 32, no. 9, pp. 499–502, May 1, 2020.

[4] X. Zhou *et al.*, "Growth, transfer printing and colour conversion techniques towards full-colour micro-LED display," *Prog. Quantum Electron.*, vol. 71, May 2020, Art. no. 100263.

[5] Z. H. Ma *et al.*, "Optical and frequency degradation behavior of GaN-based micro-LEDs for visible light communication," *Opt. Exp.*, vol. 28, no. 9, pp. 12795–12804, Apr. 2020.

[6] G. R. Lin *et al.*, "Ultrafast 2×2 green micro-LED array for optical wireless communication beyond 5 Gbit/s," *Photon. Res.*, vol. 9, no. 10, pp. 2077–2087, Oct. 2021.

[7] S. W. H. Chen *et al.*, "Full-color micro-LED display with high color stability using semipolar (20–21) InGaN LEDs and quantum-dot photoresist," *Photon. Res.*, vol. 8, no. 5, pp. 630–636, May 2020.

[8] C.-H. Ho, J. S. Speck, C. Weisbuch, and Y.-R. Wu, "Efficiency and forward voltage of blue and green lateral LEDs with V-shaped defects and random alloy fluctuation in quantum wells," *Phys. Rev. A, Gen. Phys.*, vol. 17, no. 1, Jan. 2022, Art. no. 014033.

[9] Y. Ben, F. Liang, D. Zhao, J. Yang, Z. Liu, and P. Chen, "The role of InGaN quantum barriers in improving the performance of GaN-based laser diodes," *Opt. Laser Technol.*, vol. 145, Jan. 2022, Art. no. 107523.

[10] W. Sun, S. A. Al Muyeed, R. Song, J. J. Wierer, and N. Tansu, "Integrating AlInN interlayers into InGaN/GaN multiple quantum wells for enhanced green emission," *Appl. Phys. Lett.*, vol. 112, no. 20, May 2018, Art. no. 201106.

[11] C. Jia *et al.*, "Performance improvement of GaN-based LEDs with step stage InGaN/GaN strain relief layers in GaN-based blue LEDs," *Opt. Exp.*, vol. 21, no. 7, pp. 8444–8449, Apr. 2013.

[12] H. Amano *et al.*, "Stress and defect control in GaN using low temperature interlayers," *Jpn. J. Appl. Phys.*, vol. 37, no. 12B, pp. L1540–L1542, Dec. 1998.

[13] Q. Lv *et al.*, "Realization of highly efficient InGaN green LEDs with sandwich-like multiple quantum well structure: Role of enhanced interwell carrier transport," *ACS Photon.*, vol. 6, no. 1, pp. 130–138, Jan. 2019.

[14] S. Zhu *et al.*, "Influence of quantum confined stark effect and carrier localization effect on modulation bandwidth for GaN-based LEDs," *Appl. Phys. Lett.*, vol. 111, no. 17, Oct. 2017, Art. no. 171105.

[15] J. X. Liu *et al.*, "Enhanced carrier confinement and radiative recombination in GaN-based lasers by tailoring first-barrier doping," *Opt. Exp.*, vol. 28, no. 21, pp. 32124–32131, Oct. 2020.

[16] R. Q. Wan *et al.*, "Nanohole array structured GaN-based white LEDs with improved modulation bandwidth via plasmon resonance and non-radiative energy transfer," *Photon. Res.*, vol. 9, no. 7, pp. 1213–1217, Jul. 2021.

[17] M. Funato *et al.*, "Blue, green, and amber InGaN/GaN light-emitting diodes on semipolar 11–22 GaN bulk substrates," *Jpn. J. Appl. Phys.*, vol. 45, no. No. 26, pp. L659–L662, Jul. 2006.

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